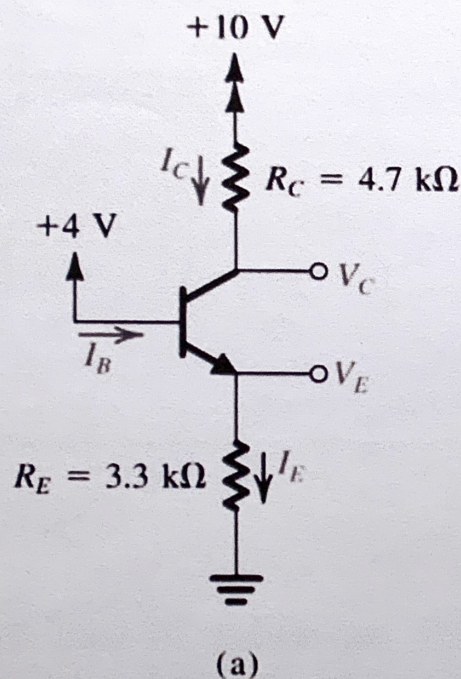


作业

D6.23 Redesign the circuit of Fig. 6.23(a) (i.e., find new values for R_E and R_C) to establish a collector current of 0.5 mA and a reverse-bias voltage on the collector-base junction of 2 V. Assume $\alpha \simeq 1$.

Ans. $R_E = 6.6 \text{ k}\Omega$; $R_C = 8 \text{ k}\Omega$



$$V_E = 4 - 0.7 = 3.3 \text{ V}$$

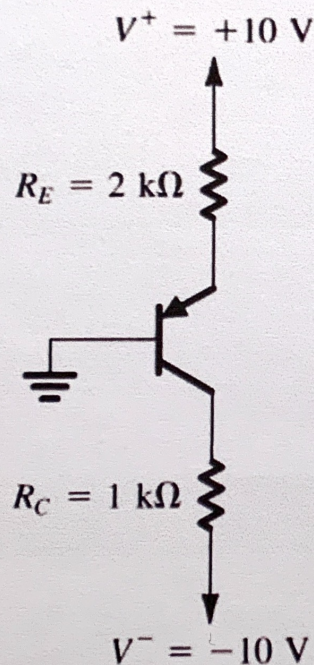
$$R_E = \frac{3.3 \text{ V}}{0.5 \text{ mA}} = 6.6 \text{ k}\Omega$$

$$V_C = 4 + 2 = 6 \text{ V}$$

$$R_C = \frac{10 - 6}{0.5 \text{ mA}} = 8 \text{ k}\Omega$$

D6.25 For the circuit in Fig. 6.26(a), find the largest value to which R_C can be raised while the transistor remains in the active mode.

Ans. 2.24 k Ω



(a)

$$V_E = 0.7 \text{ V}$$

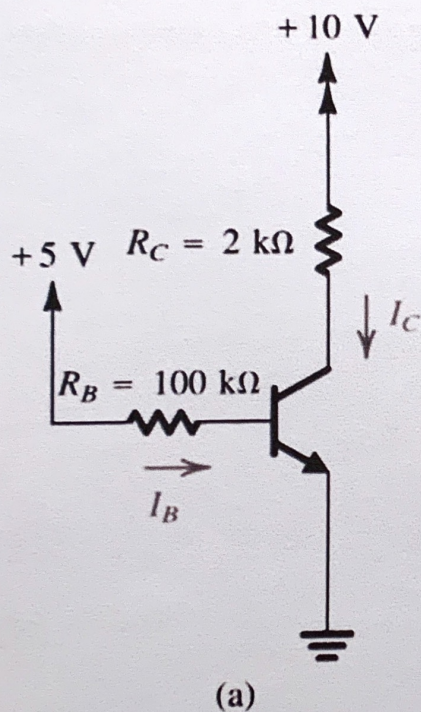
$$I_E = \frac{10 - 0.7}{2 \text{ k}} = 4.65 \text{ mA}$$

$$V_{C, \text{max}} = 0.4 \text{ V}$$

$$R_{C, \text{max}} = \frac{10.4}{4.65 \text{ mA}} = 2.24 \text{ k}\Omega$$

D6.27 The circuit of Fig. 6.27(a) is to be fabricated using a transistor type whose β is specified to be in the range of 50 to 150. That is, individual units of this same transistor type can have β values anywhere in this range. Redesign the circuit by selecting a new value for R_C so that all fabricated circuits are guaranteed to be in the active mode. What is the range of collector voltages that the fabricated circuits may exhibit?

Ans. $R_C = 1.5 \text{ k}\Omega$; $V_C = 0.3 \text{ V}$ to 6.8 V



$$I_B = \frac{5 - 0.7}{100 \text{ k}} = 43 \mu\text{A}$$

$$I_C = (50 \sim 150) \times I_B = 2.15 \text{ mA} \sim 6.45 \text{ mA}$$

I_C 越大, V_C 越小. 当 I_C 最大时仍要保证工作在放大区

工作在放大区边界时 $V_C = 0.7 - 0.4 = 0.3 \text{ V}$

$$R_C = \frac{10 - 0.3}{6.45 \text{ mA}} = 1.5 \text{ k}\Omega$$

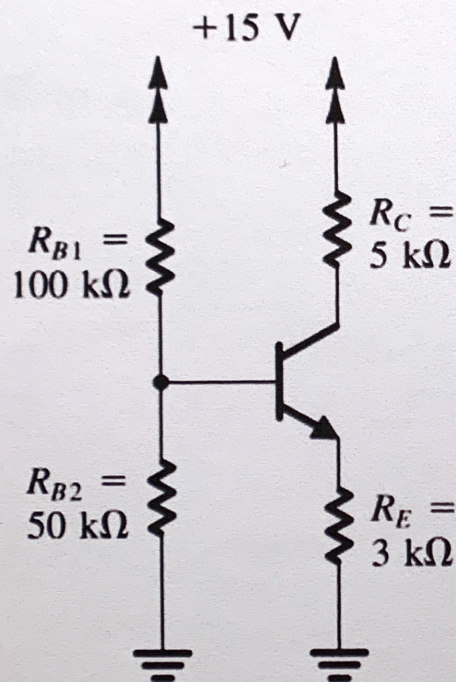
当 $I_C = 2.15 \text{ mA}$ 时 $V_C = 10 - 1.5 \text{ k}\Omega \cdot 2.15 \text{ mA} = 6.8 \text{ V}$

$$\therefore V_C = 0.3 \text{ V} \sim 6.8 \text{ V}$$

6.28 If the transistor in the circuit of Fig. 6.29(a) is replaced with another having half the value of β (i.e., $\beta = 50$), find the new value of I_C , and express the change in I_C as a percentage.

Ans. $I_C = 1.15 \text{ mA}$; -10%

$\beta = 100$ 时, $I_C = 1.28 \text{ mA}$



(a)

左边分压电路用 Thevenin 等效。

$$V_{Ther} = 15 \cdot \frac{5}{15} = 5V$$

$$R_{Ther} = 100k // 50k = 33.3k\Omega$$

$$\therefore I_B = \frac{5V - 0.7}{33.3k + 3k \times 51} = 23.08 \mu A$$

$$I_C = I_B \cdot \beta = 23.08 \mu A \times 50 = 1.15 \text{ mA}$$

$$\frac{1.15 - 1.28}{1.28} \times 100\% = -10\%$$

6.30 The circuit in Fig. E6.30 is to be connected to the circuit in Fig. 6.30(a) as indicated; specifically, the base of Q_3 is to be connected to the collector of Q_2 . If Q_3 has $\beta = 100$, find the new value of V_{C2} and the values of V_{E3} and I_{C3} .

$$\begin{aligned} V_{C,2} &= (2.75 - I_{B,3}) \times 2.7 \\ \frac{V_{C,2} - 0.7}{0.47 \times 101} &= I_{B,3} \end{aligned} \quad \Rightarrow \quad I_{B,3} = 0.134 \text{ mA}, \quad V_{C,2} = 7.06 \text{ V}$$

$$\therefore V_{E,3} = 0.47 \times I_{B,3} \times 101 = 6.36 \text{ V}$$

$$I_{C,3} = 0.134 \times 100 = 13.4 \text{ mA}$$

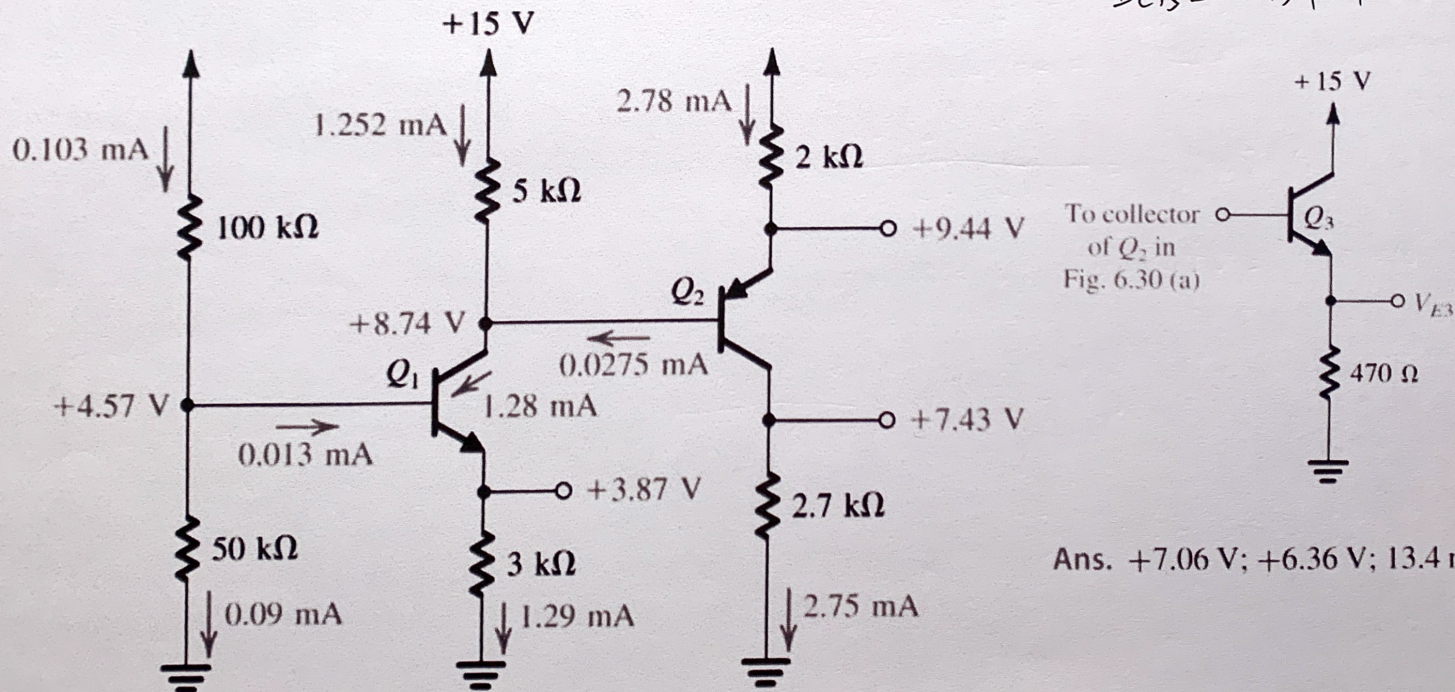


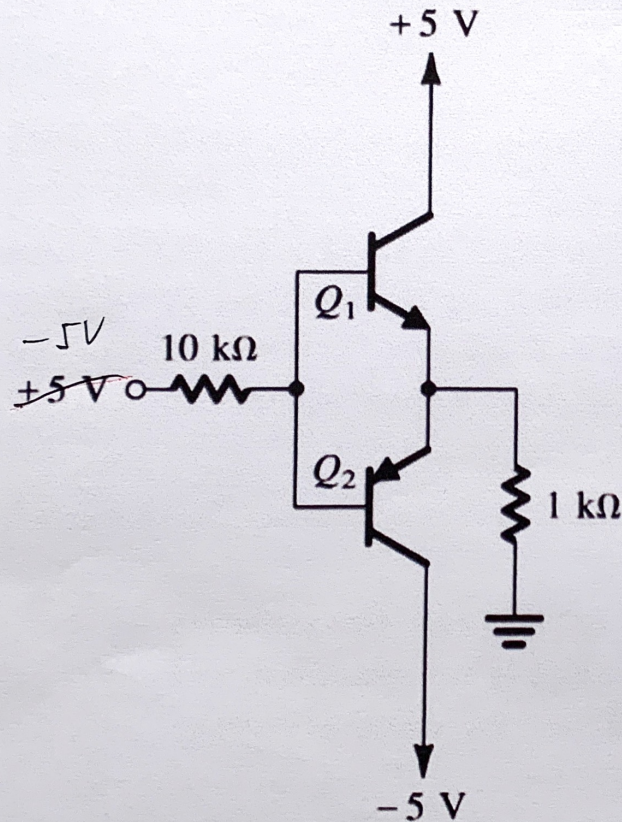
Figure E6.30

Ans. +7.06 V; +6.36 V; 13.4 mA

6.31 Solve the problem in Example 6.12 for the case of a voltage of -5 V feeding the bases. What voltage appears at the emitters? Assume $\beta = 100$.

Ans. -3.9 V

Assume $\beta = 100$.



$Q_2 \text{ on, } Q_1 \text{ off}$

$$\therefore I_{E,2} = \frac{5 - 0.7}{\frac{10}{101} + 1} = 3.91 \text{ mA}$$

$$V_{E,2} = 0 - 1\text{ k}\Omega \cdot I_{E,2} = -3.91 \text{ V}$$